

The empirical recurrence for OEIS A281799, recovered from its tail

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Abstract

OEIS A281799 records “Empirical recurrence of order 72” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1576$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 15$, so the recurrence is proved for every $n \geq 87$. Its last coefficient is $c_{72} = 64 \neq 0$, so no further tail satisfies anything shorter; any order-72 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A281799 is “Number of $n \times 5$ 0..1 arrays with no element unequal to a strict majority of its horizontal, diagonal and antidiagonal neighbors, with the exception of exactly two elements, and with new values introduced in order 0 sequentially upwards.”. It has offset 1 and begins

3, 157, 1725, 12318, 71445, 368408, 1770697, 8100180, ...

The entry states:

Empirical recurrence of order 72 (see link above)

As of the “Last modified” line on the live entry (Jan 30 2017 10:39:07, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1576$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1576$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 72: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 15$ is the first whose minimal order is exactly the stated 72.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 3152$ exact terms from $n = 15$ on, it returns order 72 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{72} c_i a(n-i),$$

c_1	15	c_{19}	−399411	c_{37}	21760608	c_{55}	−10507089
c_2	−81	c_{20}	−740271	c_{38}	−42157986	c_{56}	−3208785
c_3	158	c_{21}	86141	c_{39}	62740167	c_{57}	−5794771
c_4	57	c_{22}	335481	c_{40}	−35933508	c_{58}	−1589499
c_5	−159	c_{23}	1637718	c_{41}	16986003	c_{59}	−1930947
c_6	−1595	c_{24}	−739195	c_{42}	23590826	c_{60}	−928681
c_7	1440	c_{25}	−115965	c_{43}	−46126911	c_{61}	−408429
c_8	6552	c_{26}	−2694885	c_{44}	46609419	c_{62}	−424839
c_9	465	c_{27}	905959	c_{45}	−47043178	c_{63}	18400
c_{10}	−24399	c_{28}	4031715	c_{46}	6064062	c_{64}	−126597
c_{11}	−22932	c_{29}	−1987710	c_{47}	−16803279	c_{65}	4062
c_{12}	65122	c_{30}	492138	c_{48}	−31819297	c_{66}	−8519
c_{13}	83247	c_{31}	−5624664	c_{49}	−1542738	c_{67}	−159
c_{14}	−46326	c_{32}	3246219	c_{50}	−35858883	c_{68}	3042
c_{15}	−281577	c_{33}	−10491065	c_{51}	−5643536	c_{69}	−3471
c_{16}	−7356	c_{34}	21403605	c_{52}	−22420917	c_{70}	2604
c_{17}	401256	c_{35}	−17182452	c_{53}	−11446584	c_{71}	−720
c_{18}	323478	c_{36}	2759705	c_{54}	−8934361	c_{72}	64

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 87$, and it is the recurrence OEIS A281799 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 15$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{72} - c_1 x^{72-1} - \dots - c_{72}$. Its constant term is $-c_{72} = -64$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of

terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 72 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 72, so $\deg q = 72$, and it is monic. It holds from some index t on. From $\max(t, 15)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 72, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 72, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 64.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-72 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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